GORENSTEINNESS AT A POINT IS STABLE UNDER BASE CHANGE AND REFLECTS IT UNDER FLATNESS HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C07 (sources/stacks-duality/pinned-original-source.tex line 6614).

Allowed standard prerequisites (editorial).

- The definition of Gorensteinness at a point in terms of flatness and the fibre. Same chapter, the definition of a Gorenstein morphism preceding this lemma, invoked in the statement. Fixes the two conditions checked in each implication.
- Base change of the Gorenstein property of a ring. Same chapter, Lemma labelled lemma-gorenstein-base-change, cited at proof lines 6652-6656 and 6662-6666 and 6672-6676. Transfers Gorensteinness of the fibre in both directions.
- Flatness base change up and down. Algebra, Lemma labelled algebra-lemma-base-change-flat-up-down, cited at proof lines 6650-6652 and 6668-6672. Transfers flatness of the local ring maps in all three implications.
- Local Noetherianity of the fibres after a finite type base change. Varieties, Lemma labelled varieties-lemma-locally-Noetherian-base-change, cited at proof lines 6632-6636. Makes the statement meaningful.

Source reference. Original lemma 038R: sources/stacks-chow/tags.txt, source lines 4246–4246. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6635 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00MQ: sources/stacks-chow/tags.txt, source lines 832–832. Cross-chapter standard prerequisite located in the project tag list staged

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in the local snapshot; invoked at proof line 6651 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C03: sources/stacks-duality/pinned-original-source.tex, source lines 6480–6480. Same-chapter pointer into the author environment lemma at source line 6480; invoked at proof line 6656 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume that all the fibres X_y are locally Noetherian schemes. Let $Y' \to Y$ be locally of finite type. Let $f' : X' \to Y'$ be the base change of f. Let $x' \in X'$ be a point with image $x \in X$.*

1. *If f is Gorenstein at x, then $f' : X' \to Y'$ is Gorenstein at x'.*
2. *If f is flat at x and f' is Gorenstein at x', then f is Gorenstein at x.*
3. *If $Y' \to Y$ is flat at $f'(x')$ and f' is Gorenstein at x', then f is Gorenstein at x.*

Proof. Note that the assumption on $Y' \to Y$ implies that for $y' \in Y'$ mapping to $y \in Y$ the field extension $\kappa(y')/\kappa(y)$ is finitely generated. Hence also all the fibres $X'_{y'} = (X_y)_{\kappa(y')}$ are locally Noetherian, see Varieties, Lemma 038R. Thus the lemma makes sense. Set $y' = f'(x')$ and $y = f(x)$. Hence we get the following commutative diagram of local rings

$$\begin{array}{ccc} \mathcal{O}_{X',x'} & \longleftarrow & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_{Y',y'} & \longleftarrow & \mathcal{O}_{Y,y} \end{array}$$

where the upper left corner is a localization of the tensor product of the upper right and lower left corners over the lower right corner.

Assume f is Gorenstein at x. The flatness of $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ implies the flatness of $\mathcal{O}_{Y',y'} \to \mathcal{O}_{X',x'}$, see Algebra, Lemma 00MQ. The fact that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Gorenstein implies that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Gorenstein, see Lemma 0C03. Hence we see that f' is Gorenstein at x'.

Assume f is flat at x and f' is Gorenstein at x'. The fact that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Gorenstein implies that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Gorenstein, see Lemma 0C03. Hence we see that f is Gorenstein at x.

Assume $Y' \to Y$ is flat at y' and f' is Gorenstein at x'. The flatness of $\mathcal{O}_{Y',y'} \to \mathcal{O}_{X',x'}$ and $\mathcal{O}_{Y,y} \to \mathcal{O}_{Y',y'}$ implies the flatness of $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$, see Algebra, Lemma 00MQ. The fact that $\mathcal{O}_{X',x'}/\mathfrak{m}_{y'}\mathcal{O}_{X',x'}$ is Gorenstein implies that $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is Gorenstein, see Lemma 0C03. Hence we see that f is Gorenstein at x. □